The Geometry of Last-Layer Model Stealing

Snigdha Chandan Khilar*
snkhilar@gmail.com

June 5, 2026

Abstract

We give a geometric reading of the last-layer model-stealing attack of Carlini et al. [1], using the vocabulary of exterior differential systems (EDS) recently extended to Lie algebroids by Hohloch, Mestdag and Yasaka [2]. The set of logit vectors a transformer can emit is the common zero locus of an *ideal* with one linear part (recovered by the singular value decomposition) and one quadratic part (the ellipsoid induced by the final normalization layer). In this language the object that controls recovery of the projection matrix is the *polar space* of that quadric, which we show is exactly the tangent space of the output manifold; recovery succeeds, up to the unavoidable orthogonal gauge, precisely when a pair of *regularity* conditions—the analogues of Kähler-regularity—hold. We verify every step on a fully controlled toy model to machine precision. We then ask what lies below the last layer and report two things. First, the *intrinsic dimension* of the recoverable hidden-state manifold is an observable, invisible to the singular value decomposition and to the quadric, that detects a nonlinear sublayer and measures its effective rank. Second, we characterize what is and is not identifiable beneath the last layer, and exhibit large explicit non-identifiable fibers: different sublayers, and even different architecture widths, that produce bit-identical outputs. We are deliberately explicit about scope: the EDS framing organizes the picture but is not the engine, and the load-bearing results are classical. The contribution is a clean unified account and a concrete identifiability boundary, not a new attack.

1 Introduction

Production language models are exposed only through APIs, yet Carlini et al. [1] showed that the final *embedding projection* ("unembedding") layer of such a model can be recovered, up to symmetries, from ordinary query access. Their attack is top-down: because the last layer maps a small hidden dimension h to a large logit vector of dimension $l \gg h$, the logits live in an h-dimensional subspace, and the singular value decomposition (SVD) of enough query responses reveals both h and the projection matrix up to a linear change of basis. A refinement (their Appendix H) exploits the fact that the final normalization layer places hidden states on a sphere, so the logits lie on an ellipsoid; fitting that ellipsoid sharpens the recovery from "up to an invertible matrix" to "up to an orthogonal matrix."

This note makes two contributions, both modest and clearly delimited.

A geometric account. We recast the attack in the language of exterior differential systems, which is the natural setting for "reconstruct a global object from local data under constraints, modulo a

*Independent Researcher.

symmetry group." The attainable logits form the integral variety of an *ideal*; the object governing recovery of the projection is the *polar space* of its quadratic generator; and recovery is well posed exactly under regularity conditions that are the affine analogues of the Kähler-regularity used in the Cartan–Kähler theorem [2,3]. The single-layer case turns out to be Frobenius-integrable—which is precisely why the attack is closed-form rather than iterative—and we confirm the whole picture numerically.

An identifiability wall. We then look one layer deeper. We observe that the *intrinsic dimension* of the recoverable hidden-state manifold is an extraction observable distinct from the linear span the SVD measures: when a low-rank nonlinear sublayer is present, the span overstates the content dimension, and the intrinsic dimension reveals the bottleneck. Finally we give a crisp identifiability characterization of the sublayer and demonstrate, with machine-precision examples, that most of its parameters lie in a non-identifiable fiber. This explains mechanically why the attack has not been extended past one layer: it is not a missing trick but a property of the observation map.

Honesty about scope. The geometric language is organizing, not enabling: at no point does it produce a result the standard linear-algebra and manifold tools could not. The identifiability statements rest on classical neural-network identifiability [4] and on the known fact that learned representations have low intrinsic dimension [5]. We state this plainly so the note is read as a unified exposition with a concrete identifiability boundary, not as a new attack.

Roadmap. Section 2 is a self-contained primer on both halves of the story—model stealing and the handful of differential-geometry notions we borrow—written for readers who know neither; specialists can skip to Section 3. Sections 4–6 develop and verify the single-layer picture, and Sections 7–8 look beneath the last layer.

2 Background and intuition

This section assumes no prior exposure to either model extraction or exterior differential systems (EDS). Experts may skip to Section 3.

What "model stealing" means. A language model is usually served behind an API: you send text and receive, for each possible next token, a score (a *logit*) that the model turns into a probability. The provider keeps the model's weights secret. *Model stealing* asks how much of those weights an outsider can reconstruct using only API queries. One does not expect to copy a multi-billion-parameter model from query access; the surprising result of [1] is that one specific piece—the final linear layer—can be recovered exactly, up to an unavoidable ambiguity, and cheaply.

Why the last layer is the easy target. A transformer carries information in a vector of width h (the "hidden" or "residual" dimension), then multiplies that vector by a matrix $\mathbf{W}$ to produce one logit per vocabulary token. The vocabulary is large (l in the tens of thousands) while h is comparatively small, so $\mathbf{W}$ is a tall, thin, rank-h matrix: it maps a small space up into a large one. That gap, $h \ll l$, is the crack the attack pries open.

The rank trick. Query the model on many different prompts and stack the logit vectors as columns of a matrix. Although each column lives in l dimensions, every column is $\mathbf{W}$ times something h-dimensional, so all of them lie in the same h-dimensional subspace. Once you have queried more than h times, new responses become linear combinations of old ones. The singular value decomposition (SVD) detects this: it reports exactly h large singular values and a sharp drop afterwards. Counting them recovers the hidden width (Figure 1); a little more linear algebra recovers $\mathbf{W}$ itself, up to a change of basis.

From a sphere to an ellipsoid. Modern transformers *normalize* the hidden vector just before the last layer, which forces it to have fixed length—it lives on a sphere. A linear map sends a sphere to an *ellipsoid*. So the logits do not merely fill an h-dimensional subspace; they lie on an ellipsoidal surface inside it. Fitting that ellipsoid pins down more of $\mathbf{W}$: it sharpens "known up to any invertible change of basis" to "known up to a rotation." The leftover rotation is genuinely unrecoverable, for a simple reason given in Section 5.

Three borrowed ideas. The EDS vocabulary we use names three things that are already implicitly present above.

- An *ideal* is just the collection of equations every observation satisfies. Here there are two kinds: linear ones (the logits lie in the subspace) and one quadratic one (they lie on the ellipsoid). The surface they cut out is the *integral variety*—the set of attainable outputs.
- A *polar space* answers "given part of a solution, which directions can extend it?" For a quadric this is the classical notion of points *conjugate* with respect to the surface, and—as we show—it is exactly the tangent plane to the ellipsoid. Recovering the last layer amounts to recovering this field of tangent planes.
- *Regularity* is the package of nondegeneracy conditions that make the reconstruction unique and stable: a clean gap in the spectrum, and a genuinely curved (nondegenerate) ellipsoid. When they fail, the attack fails in a predictable way.

Why bring in EDS at all? Honestly, for the single layer it is a unifying language rather than a new tool: rank recovery, ellipsoid recovery, the rotation ambiguity, and the stability conditions become one object with three features. Its real payoff is conceptual—it tells us in advance (Remark 1) why the single-layer attack is one-shot, and it frames the genuinely hard question, "what can be learned about the layer underneath," as a question about the geometry of a curved surface (Sections 7–8).

3 Setup

Let $\mathcal{X}$ be the token vocabulary, $|\mathcal{X}| = l$. A model produces $f_\theta(p) = \mathrm{softmax}(\mathbf{W}\, g_\theta(p))$, where $g_\theta \colon \mathcal{X}^N \to \mathbb{R}^h$ computes a hidden state and $\mathbf{W} \in \mathbb{R}^{l\times h}$ is the projection, with $h \ll l$. We assume the idealized oracle that returns the full logit vector $z = \mathbf{W}\, g_\theta(p) \in \mathbb{R}^l$; the engineering needed to recover logits from top-K log-probabilities and a logit bias is treated at length in [1] and is orthogonal to what follows. We assume the final block is a normalization (RMSNorm or LayerNorm) followed by $\mathbf{W}$, so the attainable hidden states lie on a sphere $S \subseteq \mathbb{R}^h$ and the logits lie in $V := \mathrm{col}(\mathbf{W})$.

4 The output ideal

The attainable logits are the common zeros of two families of constraints.

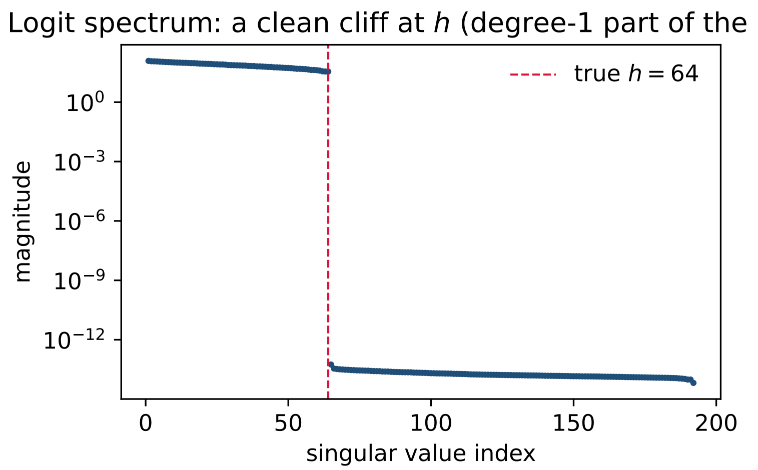

Figure 1: The degree-1 part of the ideal. On a toy model with $h = 64$, the logit singular spectrum drops by fourteen orders of magnitude at index h: the recovered $\hat{h}$ is exactly 64.

Degree-1 generators (rank). Let $\{\nu_1, \dots, \nu_{l-h}\}$ span $V^{\perp}$. Each linear form $\ell_a(z) = \langle \nu_a, z \rangle$ vanishes on every response; these are what the SVD recovers as the directions with zero singular value.

Degree-2 generator (normalization). Because $\|g\|$ is fixed by normalization, $z = \mathbf{W}g$ satisfies a single quadric $q(z) = z^{\top}\hat{A}z - 1 = 0$, where $\hat{A}$ is symmetric positive semidefinite of rank h with $\ker \hat{A} = V^{\perp}$. With $U \in \mathbb{R}^{l \times h}$ an orthonormal basis of V and $x = U^{\top}z$, this reads $x^{\top}Ax = 1$ for a positive definite $A \in \mathbb{R}^{h \times h}$, and the recovery factorization $\mathbf{W} = UM^{-1}O$ with $A = M^{\top}M$ and O orthogonal follows [1].

Thus the ideal is $\mathcal{I} = \langle \ell_1, \dots, \ell_{l-h},\ q \rangle$ and the output manifold is the ellipsoid $\mathcal{M} = \{\ell_a = 0,\ q = 1\}$. Two structural facts are worth recording.

Remark 1 (Frobenius triviality)**.** *Passing to the exterior ideal generated by differentials, the generators are the constant covectors ν_a and the one-form $dq = 2\hat{A}z$. All are closed, so $\mathcal{I}$ is a differential ideal generated by 1-forms: the system is Frobenius-integrable. This is the structural reason the attack is one-shot linear algebra with no integrability obstruction, and it tells us in advance that the heavier Cartan–Kähler machinery can only become necessary when more than one layer is involved (Section 7).*

Remark 2 (Architecture fingerprint)**.** *The count of degree-1 generators distinguishes normalizations: LayerNorm centers before normalizing, adding one linear generator and dropping the effective dimension by one, which is the singular-value signal used in [1] to tell LayerNorm from RMSNorm.*

5 Polar space, gauge, and recovery

In the Cartan–Kähler theory the *polar space* of an integral element controls which directions extend it [2,3]. Here the governing generator is the quadric, so the relevant object is the classical polar (conjugate) space of that quadric.

Proposition 1 (Polar space is the tangent space)**.** *At a point $z_0 \in \mathcal{M}$ the polar space $H(z_0) = (\hat{A}z_0)^\perp$ equals the tangent space $T_{z_0}\mathcal{M}$. Consequently recovering the layer is recovering the field of polar hyperplanes, each tangency condition $z_0^\top \hat{A} v = 0$ being one linear equation in the entries of $\hat{A}$.*

We verify this independently: computing a tangent direction of the output manifold by finite differences (without using the recovered $\hat{A}$), its cosine with the recovered polar normal is 5.8×10^{-7}, limited only by the finite-difference step.

The gauge. The symmetric form $\hat{A} = M^\top M$ fixes only the Gram part of $\mathbf{W}$; the antisymmetric complement, of dimension $\binom{h}{2}$, is free. This is the orthogonal gauge: under $\mathbf{W} \mapsto \mathbf{W}O^\top$, $g \mapsto Og$, the logits $z = \mathbf{W}O^\top Og = \mathbf{W}g$ are unchanged for every prompt, so the entire output distribution is $O(h)$-invariant and no statistic of the logits can distinguish points of the orbit. Recovery up to $O(h)$ is therefore information-theoretically optimal from logits alone.

On the toy model ($h = 64$), Cholesky followed by a *scaled-orthogonal* alignment reconstructs $\mathbf{W}$ to root-mean-square error 6×10^{-16}—the same machine precision as a full affine alignment with h^2 free parameters—while the aligning rotation has $\|\Omega - I\|_F \approx 11.5 \approx \sqrt{2h}$, i.e. a generic rotation. The quadric thus resolves the symmetric content exactly and leaves precisely the $O(h)$ gauge, as predicted.

6 Regularity, and what breaks it

Recovery is well posed under the affine analogues of Kähler-regularity:

(R1) Spectral gap. A strictly positive multiplicative gap between σ_h and σ_{h+1}, equivalently full rank h of both the hidden-state matrix and $\mathbf{W}$.

(R2) Nondegenerate quadric. $\hat{A}$ positive definite of rank h on V; equivalently the activations are not confined to a proper sub-variety, so the ellipsoid-fitting system has full rank.

(R3) Uniformity. (R1)–(R2) hold on a neighborhood; automatic once $\hat{A} \succ 0$.

These are not decorative. Table 1 sweeps i.i.d. logit noise (the defense of [1, App. I]) and shows two distinct laws: the rank gap degrades like $1/\sigma$ but stays above 1 even at $\sigma = 1$, so $\hat{h}$ is recovered at every level—*rank is robust*; while the orthogonal-recovery error grows linearly, RMS $\approx 0.036\,\sigma$—*the projection is fragile* and needs the regularity to hold tightly. Confining activations to an effective-rank subspace (an R1 violation) makes the attack return the effective rank rather than the nominal width—reproducing the GPT-2-Small anomaly of [1], where 757 was recovered for a 768-dimensional model. Figure 2 (left) plots both laws.

7 Below the last layer

After the single-layer attack, the hidden states $g(p)$ are themselves known up to the global rotation. We now ask what this reveals about the block beneath. Consider a toy with k-dimensional content fed through an MLP block with a residual connection,

$$g = \mathrm{norm}\big(x + \mathbf{W}_2\, \phi(\mathbf{W}_1 x)\big), \qquad x = Bs,\ s \in \mathbb{R}^k,$$

with $\phi = \tanh$, hidden width m, residual width h. The attainable hidden states now lie on a *curved* k-dimensional submanifold of the sphere whose linear span can be far larger than k. The intuition

Table 1: Noise sweep on the toy model ($h = 64$, unit-scale logits). Rank recovery (R1) is robust; projection recovery (R2/R3) degrades linearly.

σ	$\hat{h}$	gap σ_h/σ_{h+1}	orthogonal RMS
10^{-4}	64	2.48×10^4	3.0×10^{-6}
10^{-3}	64	2.49×10^3	3.2×10^{-5}
10^{-2}	64	2.48×10^2	3.0×10^{-4}
10^{-1}	64	2.46×10^1	2.9×10^{-3}
1.0	64	2.56×10^0	3.6×10^{-2}

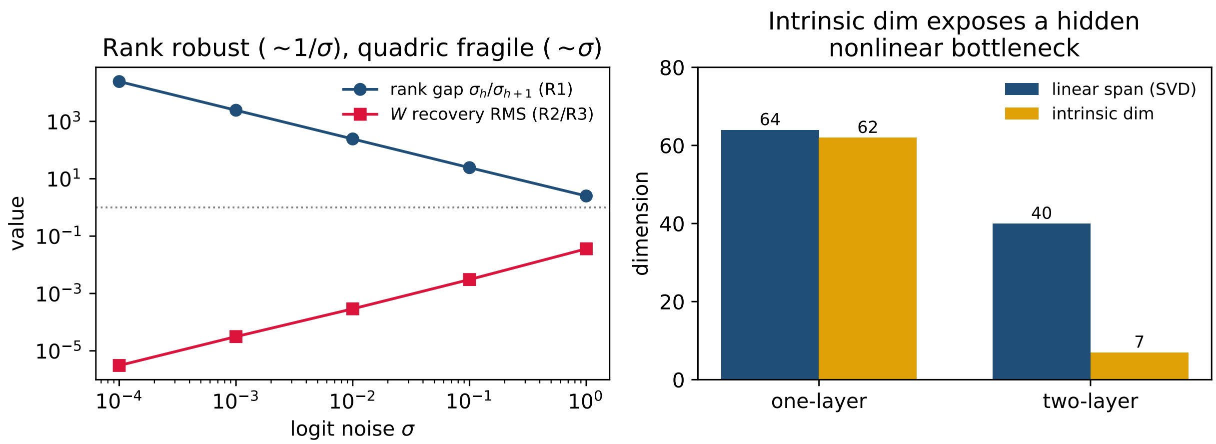

Figure 2: **Left:** regularity is load-bearing. Under logit noise the rank gap (R1) decays like $1/\sigma$ but never falls below 1; the projection recovery error (R2/R3) grows linearly. **Right:** below the last layer, the linear span (what the SVD reports) overstates the content; the intrinsic manifold dimension recovers it and exposes the nonlinear bottleneck.

is the same as a circle in the plane: a circle is a one-dimensional object, yet it does not fit in any single line—its linear span is two-dimensional. A curved k-dimensional manifold likewise needs more than k linear dimensions to contain it, and the nonlinearity ϕ is precisely what bends the content manifold so that its span inflates above its true dimension.

An observable the linear attack misses. The SVD sees only the linear span. On a two-layer toy with $h = 64$, $k = 8$, $m = 32$ the span is 40, the quadric is positive definite and well posed (design rank 820/820), and the attack returns a clean rank-40 projection with no internal sign that anything is amiss—it reports a featureless 40-dimensional linear layer. But the *intrinsic dimension* of the recovered manifold, estimated by local PCA, is 7–8: the true content dimension. The gap between span (40) and intrinsic dimension (8) is the fingerprint of a low-rank nonlinear bottleneck. The one-layer control gives intrinsic dimension 62 against span 64, i.e. span $- 1$ (the sphere), with no gap. In the EDS reading the intrinsic dimension is the first Cartan character and the span–intrinsic gap is the osculating (second-order) data the linear attack discards. Figure 2 (right) shows the contrast.

8 The identifiability wall

The observable is the hidden-state manifold $\mathcal{S}$ up to the global orthogonal gauge. We characterize the sublayer $x \mapsto x + \mathbf{W}_2\phi(\mathbf{W}_1 x)$ relative to it. The governing intuition is simple: one can only learn about parts of a network that the inputs actually exercise, and only up to relabelings that leave every output unchanged. Both effects turn out to be large here.

Proposition 2 (Identifiability boundary, toy model)**.** *From $\mathcal{S}$ the sublayer is determined only by its input–output behaviour on the input support $\mathcal{X} = \operatorname{col}(B)$, and only up to: the global rotation $O(h)$; input reparametrization $GL(k)$ on $\mathcal{X}$; and, for minimal* tanh *realizations, neuron sign-permutation [4]. The following are free fibers, hence unrecoverable:*

(a) *the action of $\mathbf{W}_1$ on $\mathcal{X}^\perp$;*

(b) *the MLP width m (only the minimal realization on $\mathcal{X}$ is pinned);*

(c) *B beyond its column space (a full $GL(k)$);*

(d) *the parametrization of the input distribution—only its support matters.*

We make the fibers concrete with bit-level examples (all logit differences below are at machine precision, $\sim 10^{-14}$).

(a) Off-support freedom. Let $\Delta = \Delta(I - P_{\mathcal{X}})$ act only on $\mathcal{X}^\perp$. Then $\mathbf{W}_1$ and $\mathbf{W}_1 + \Delta$ produce identical outputs. For $h = 64$, $k = 8$, $m = 32$ this hides $m(h-k) = 1792$ of the 2048 parameters of $\mathbf{W}_1$ (87.5%), with $\|\Delta\|_F \approx 42$.

(b) Width. Appending a cancelling neuron pair ($\mathbf{W}_1 \to \left(\begin{smallmatrix}\mathbf{W}_1\\ w\\ w\end{smallmatrix}\right)$, $\mathbf{W}_2 \to [\mathbf{W}_2\ c\ {-c}]$) changes the width from $m = 32$ to 34 while leaving every output unchanged.

(c) Input reparametrization. Replacing B by BR for $R \in GL(k)$ leaves the attainable input set, hence $\mathcal{S}$, unchanged.

The interpretation is sharp: the part of the sublayer one would most want, the full $\mathbf{W}_1$, lies almost entirely inside the fiber. This is the mechanical reason the attack stops at one layer—the observation map has a large kernel below the last linear layer—and it reframes the open "extend beyond one layer" problem of [1] as an identifiability question rather than an algorithmic one.

9 Scope, novelty, and related work

We are explicit about what is and is not new. *What is reproduced:* the last-layer recovery, including recovery up to an orthogonal matrix via the normalization-induced ellipsoid, is from [1]. *What the geometric language adds:* a single account in which rank recovery, ellipsoid recovery, the $O(h)$ gauge, and the regularity conditions are one structure—the ideal, its polar space, and its Kähler-regularity—together with the Frobenius observation (Remark 1) that explains why the single-layer attack is closed-form. We stress that this framing is organizing, not enabling. *What is a genuinely new observation, though built on known tools:* the intrinsic dimension of the recoverable manifold as an extraction observable (Section 7), and the explicit identifiability boundary with its fibers (Section 8). The first leans on the established fact that representations have low intrinsic dimension [5]; the second on classical network identifiability [4]. Concurrent and prior work on the same attack family includes [6,7]. The EDS toolkit we borrow from is that of [3] and its Lie-algebroid extension [2].

We did *not* recover the sublayer parameters, and Section 8 indicates this is blocked by identifiability rather than by a missing technique. A clean impossibility theorem for realistic architectures, with a matching positive recovery result on the identifiable quotient, is the natural next target and is left open.

10 Conclusion

Last-layer model stealing has a tidy geometric description: the logits trace the integral variety of an ideal with a linear and a quadratic generator, the quadric's polar space is the manifold's tangent space and controls recovery, and the attack succeeds up to the orthogonal gauge exactly under regularity conditions that we showed are load-bearing. One layer down, the linear span the SVD reports can hide a low-rank nonlinear bottleneck that the intrinsic dimension reveals, and the sublayer's parameters live largely in an explicit non-identifiable fiber. The geometry is a clarifying lens; the wall beneath the last layer is real and, we argue, the more fruitful object of study.

Reproducibility. Every number in this note is produced by a small NumPy library and a single reproduction script; figures are regenerated by one command. Code: `https://github.com/nssprogrammer/eds-stealing`.

References

[1] N. Carlini et al., *Stealing Part of a Production Language Model.* ICML 2024. arXiv:2403.06634.

[2] S. Hohloch, T. Mestdag, K. Yasaka, *The Cartan–Kähler theorem for exterior differential systems on transitive Lie algebroids.* arXiv:2605.29083 (2026).

[3] R. L. Bryant, S. S. Chern, R. B. Gardner, H. L. Goldschmidt, P. A. Griffiths, *Exterior Differential Systems.* Springer, 1991.

[4] H. J. Sussmann, *Uniqueness of the weights for minimal feedforward nets with a given input–output map.* Neural Networks 5(4):589–593, 1992.

[5] A. Ansuini, A. Laio, J. H. Macke, D. Zoccolan, *Intrinsic dimension of data representations in deep neural networks.* NeurIPS 2019. arXiv:1905.12784.

[6] M. Finlayson, S. Swayamdipta, X. Ren, *Logits of API-protected LLMs leak proprietary information.* arXiv:2403.09539 (2024).

[7] S. Zanella-Béguelin, S. Tople, A. Paverd, B. Köpf, *Grey-box extraction of natural language models.* ICML 2021.